

Knowledge Map

What I Know, What's Known, and What's Open

Patrick S. Mahon

Personal reference document — not for submission

May 2026

Abstract

A consolidated reference covering everything explored in this research programme: what I understand, what is known (and by whom), what I proved that turned out to be classical, what remains genuinely open, and where the understanding might lead. Not a paper. A map for finding the next problem.

Contents

1	The central question	2
2	Layer 1: From charges to measures	2
2.1	The classical results (all known)	2
2.2	What I proved (turned out classical)	2
2.3	What I understand from this	3
3	Layer 2: From dual space to realization	3
3.1	The classical results (all known)	3
3.2	What I proved (turned out classical)	3
3.3	What I understand from this	3
4	Layer 3: From probability to value-definiteness	4
4.1	The classical results (all known)	4
4.2	What I proved (classical + one novel framing)	4
4.3	What I understand from this	4
5	Delay reconstruction	5
5.1	The classical results	5
5.2	What I proved (mostly classical)	5
5.3	What is genuinely novel (but motivation unclear)	5
5.4	What I understand from this	6
6	Dead ends and lessons	6
7	What remains genuinely open	7
8	The understanding, condensed	7
9	References by subject	8

1 The central question

When does observationally coherent data determine a genuine probability measure, and what additional commitments does this require?

This decomposes into three layers:

1. **Extension:** compatible finitely additive charges on a directed system of Boolean algebras $\rightarrow$ σ -additive probability measure.
2. **Realization:** measure on the dual space (Stone space) $\rightarrow$ measure on an underlying state space Ω .
3. **Value-definiteness:** probabilistic description $\rightarrow$ simultaneous definite values for all observables.

Each layer requires a commitment the previous does not. The algebraic structure of the observation algebra constrains which commitments are available; distributivity is sufficient to make all three available.

2 Layer 1: From charges to measures

2.1 The classical results (all known)

- **Continuity from above:** A finitely additive charge on an algebra is σ -additive iff $E_n \downarrow \emptyset \Rightarrow \ell(E_n) \rightarrow 0$ [17, §13, Theorem D].
- **Carathéodory extension:** A σ -additive premeasure on a semiring extends uniquely to a measure on the generated σ -algebra [17, §13], [5, Vol. 1, Ch. 1].
- **Kolmogorov extension:** Compatible finite-dimensional distributions on a product space extend to a unique probability measure, under standard Borel / tightness conditions [4, Ch. 36].
- **Choksi inverse limits:** Compatible measures on an inverse system of compact Hausdorff spaces extend to the inverse limit [9, Theorem 1].
- **Yosida–Hewitt decomposition:** Every bounded finitely additive charge decomposes uniquely as $\ell = \ell_c + \ell_p$ where ℓ_c is σ -additive and ℓ_p is purely finitely additive [41].

2.2 What I proved (turned out classical)

Collective exhaustion (CE) in a directed system: defined as $\exists j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \rightarrow 0$. Under compatibility, the existential witness is vacuous: $\ell_j(\pi_{ij}^{-1}(E_n)) = \ell_i(E_n)$ for all $j \geq i$. So CE = σ -additivity of each ℓ_i . This is not a theorem; it is the observation that the definition contains a redundant quantifier.

The directed-system extension theorem (Paper I, Theorem 2.3) is Kolmogorov extension in directed-system language with “sequential common refinements” replacing the usual product structure.

The Lean formalization (0 sorry on the Carathéodory route, 1 sorry for Yosida–Hewitt on the Stone route) is genuine verified infrastructure, even though the theorems are classical.

2.3 What I understand from this

1. The passage from finite additivity to σ -additivity is the critical non-finitary step in probability. Compatibility, normalization, and finite additivity are all first-order / finitary conditions.
2. No first-order condition can force σ -additivity. Immediate from Łoś's theorem: the ultraproduct of Dirac masses on $\mathbb{N}$ is a purely finitely additive charge [20, 12].
3. The finite-cofinite charge on $\mathbb{N}$ is the canonical obstruction: normalized, finitely additive, not σ -additive, and arises as the ultraproduct witness.
4. The Yosida–Hewitt decomposition gives the exact characterization: $\text{CE} \Leftrightarrow \ell_p = 0$.

3 Layer 2: From dual space to realization

3.1 The classical results (all known)

- **Stone representation:** Every Boolean algebra is isomorphic to the clopen algebra of a compact, Hausdorff, totally disconnected space (its Stone space) [22].
- **Charge $\leftrightarrow$ measure on Stone space:** Every finitely additive charge on a Boolean algebra B corresponds to a regular Borel (Baire) measure on its Stone space $\text{St}(B)$ (the space of ultrafilters on B) [13, §326], [17, §53].
- **σ -additivity $\leftrightarrow$ support:** A charge on B is σ -additive iff the corresponding Stone-space measure vanishes on meagre Baire sets, equivalently assigns full mass to countably complete ultrafilters [32, Theorem 10.5.3].
- **Principal ultrafilters:** Every principal ultrafilter is countably complete. The converse holds for $\mathcal{P}(\kappa)$ when κ is below the first measurable cardinal (Ulam's theorem). For non- σ -complete algebras (typical cylinder algebras), countably-complete non-principal ultrafilters can exist without large cardinals.

3.2 What I proved (turned out classical)

Stone measure exists unconditionally: Any compatible charges produce a Baire probability measure $\hat{\mu}$ on $\text{St}(\mathcal{C})$. This is Stone representation + Choksi. Lean-verified (0 sorry).

Descent requires $\hat{\mu}(\text{pure}(\Omega)) = 1$ (where $\text{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$ is the principal ultrafilter at ω): concentration on principal ultrafilters. Under discriminability and the assumption that all countably-complete ultrafilters in the relevant subalgebra are principal (which holds for σ -complete algebras below the first measurable cardinal), this is equivalent to σ -additivity. The support characterization is Rao–Rao 10.5.3; the principal-vs-countably-complete gap is the subtlety.

Realization is unconstrained: Any surjective-projecting $S \subseteq \text{St}(\mathcal{C})$ can serve as Ω . This is the abstractness of Boolean algebras recognized since Stone (1936).

3.3 What I understand from this

1. The Stone construction gives the *unconditional* object: $\hat{\mu}$ always exists. It is the “empirically adequate” object in van Fraassen’s sense.
2. Descent to Ω is a *choice*: which ultrafilters count as “realized” is additional structure over the algebra.

3. The three-layer nesting:

$$\text{pure}(\Omega) \subseteq \{\text{countably complete ultrafilters}\} \subseteq \text{St}(\mathcal{C}).$$

σ -additivity gives mass on the middle set (intrinsic). Descent needs mass on the inner set (depends on Ω).

4. The van Fraassen identification (Stone measure = empirical adequacy (EA), descent = realist commitment) has no precedent in the literature. It is a philosophical contribution, not a mathematical one.

4 Layer 3: From probability to value-definiteness

4.1 The classical results (all known)

- **Boolean algebras admit 2-valued homomorphisms:** Ultrafilters on Boolean algebras are exactly the 2-valued homomorphisms. Existence by Zorn. *Stone (1936); every Boolean algebra textbook.*
- **Kochen–Specker:** For the lattice $L(H)$ of closed linear subspaces of a Hilbert space H with $\dim H \geq 3$, no dispersion-free state (2-valued homomorphism) exists [25].
- **Gleason:** For $L(H)$, $\dim H \geq 3$, every σ -additive state is $s(P) = \text{tr}(\rho P)$ [15].
- **Bub–Clifton:** Given a state ρ and a preferred observable R , there is a unique maximal Boolean subalgebra $D(\rho, R) \subseteq L(H)$ on which the state is a mixture of dispersion-free states. [7, 18].
- **Orthomodular lattice (OML) duality:** The category of OMLs is dually equivalent to a category of orthomodular spaces. Principal filters $P(A)$ are part of the dual structure (unlike Stone duality where they are invisible). [29]. Zero citations in quantum foundations as of 2026.
- **Colimit generation:** Gluing Boolean algebras via categorical pushout produces orthomodular lattices, generically non-distributive. [16].

4.2 What I proved (classical + one novel framing)

Commensurability theorem: Distributivity is sufficient for value-definite realism (VDR); for $L(H)$, $\dim H \geq 3$, non-distributivity blocks it. Lean-verified (0 sorry, 1 axiom for Kochen–Specker (KS)). *The mathematics is the assembly of Stone + KS + Gleason + Bub–Clifton. Not a new theorem.*

The EA / probabilistic realism (PR) / VDR vocabulary connecting van Fraassen’s empirical adequacy to quantum logic via McDonald–Bimbó duality is novel framing. No precedent found.

The coherence filtration $\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{\text{df}}(A)$ with three named transitions (pasting, regularity, sharpness) organizes known results. The chain itself is textbook (Pták–Pulmannová, Kalmbach); the transition names and the observation that distributivity collapses the filtration are new packaging.

4.3 What I understand from this

1. **Distributivity controls value-definiteness.** Boolean algebras have ultrafilters (VDR available). Non-Boolean OMLs ($\dim \geq 3$) do not (KS). The dividing line is distributivity — sufficient but not necessary in full generality ($\dim 2$, horizontal sums).

2. **The algebra can constrain realization in the OML case.** Unlike Boolean Stone duality, the McDonald–Bimbó dual carries $P(A)$ as structure. “Realization is additional structure” in the Boolean case; it is *constrained* structure in the OML case.
3. **The realism debate has different mathematical answers for different algebras.** Boolean: EA, PR, VDR all available; philosophical choice. OML ($\dim \geq 3$): VDR blocked; PR available (Gleason); EA always available.
4. **The topos programme is more powerful.** Isham–Butterfield / Döring–Isham / Heunen–Landsman–Spitters encode the same structure via spectral presheaves and Bohri-fication. Our approach is coarser (single dual space, not presheaf over contexts) but connects more directly to van Fraassen.

5 Delay reconstruction

5.1 The classical results

- **Takens:** Generic smooth observable gives an injective delay map for $L \geq 2 \dim X$ [39].
- **Sauer–Yorke–Casdagli:** Prevalence-based embedding theorem [35].
- **Generating partitions:** $h(T, P) = h(T)$ iff P generates [37, 26].
- **Grassberger–Procaccia:** Correlation integral estimates D_2 and K_2 (Rényi-2 dimension and entropy) from delay data.
- **False nearest neighbours (FNN):** Embedding dimension selection [24]. Sensitive to noise [33].
- **Ragwitz–Kantz:** Local prediction error minimization for (d, L) selection [31].
- **Botvinick-Greenhouse et al.:** Measure-theoretic Takens via optimal transport [6].

5.2 What I proved (mostly classical)

Reconstruction theorem: Observable σ -algebra = full σ -algebra mod μ iff delay algebra dense in L^2 iff delay map is measure-theoretic embedding. This is Krieger’s generating partition theorem [26] in delay-map language.

Predictive factorization, Chapman–Kolmogorov, Koopman–Perron: All classical (Rokhlin, Kolmogorov, Koopman 1931).

Finite-sample convergence rates: Minimax rates $n^{-s/(2s+d)}$ [38]; elbow stopping via Lepski [27]. Not cited in the original work — a significant attribution gap.

5.3 What is genuinely novel (but motivation unclear)

Geometric $\neq$ algebraic reconstruction (2026-05-14): The bridge note claimed $\delta(L) \rightarrow 0 \Leftrightarrow (\mu \otimes \mu)(R_L) \rightarrow 0$ under a fibre mixing condition. This is **false**: the core identity (Lemma 2.1 of the bridge note) has a disintegration error. The conditional variance $\mathbb{E}[\text{Var}(\mathbf{1}_S \mid \mathcal{O}_L)] = \sum_z p_z \mu_z(S) \mu_z(S^c)$ is weighted by p_z , while the collision integral $(1/2) \int_{R_L} |\mathbf{1}_S - \mathbf{1}_{S'}|^2 d(\mu \otimes \mu) = \sum_z p_z^2 \mu_z(S) \mu_z(S^c)$ is weighted by p_z^2 . The bridge note equated them; they are genuinely different.

Skew-product counterexample: $X = A^{\mathbb{Z}} \times B^{\mathbb{Z}}$, $T = \sigma \times \sigma$, $h(a, b) = a_0$, independent iid uniform. Then collision = $2^{-(L+1)} \rightarrow 0$ (fibres shrink in the a -coordinate) but $\delta_L = 1/4$ for all L (the event $\{b_0 = 0\}$ is maximally unresolved). This falsifies even the “easy direction.”

What survives: The definitions (collision probability, Rokhlin distance, fibre mixing) are well-posed. The trivial direction $\delta \rightarrow 0 \Rightarrow \text{collision} \rightarrow 0$ holds. The converse and the entropy characterization $\delta \rightarrow 0 \Leftrightarrow H_2 \rightarrow \infty$ are both dead.

Replacement direction: A short note proving that geometric and algebraic reconstruction are inequivalent, with the skew-product as the central theorem. The open question becomes: what conditions on the factor map make $\text{collision} \rightarrow 0$ imply $\delta \rightarrow 0$?

5.4 What I understand from this

1. The reconstruction problem is *algebraically underdetermined*: the observation algebra does not constrain the realization. Embedding selection (d, L) is necessarily empirical.
2. Every practical reconstruction criterion (FNN, mutual information, Lyapunov convergence, prediction error) implicitly assumes a modelling commitment. Deterministic criteria fail under noise because noise violates the commitment.
3. The Rokhlin distance $\delta(\mathcal{O})$ is the measure-theoretic notion of “how close is the observable σ -algebra to generating everything.” It is the Rokhlin metric, named and studied since 1967.
4. Collision probability / Rényi-2 entropy measures the “collision structure” of the fibre decomposition. Grassberger–Procaccia already extracts this from data.
5. The bridge theorem linking collision convergence to σ -algebra generation is false: the disintegration error (p_z vs p_z^2 weighting) means the proposed identity is simply the wrong identity, and the inequivalence of geometric and algebraic reconstruction is known/obvious.

6 Dead ends and lessons

1. **De Finetti via Stone:** Exchangeability forces σ -additivity (Hausdorff moments). The “exotic part” of the Stone measure is empty. Dead end.
2. **Stationary Stone:** The Yosida–Hewitt decomposition commutes with shift-invariance. An observation, not a theorem. The real open question (ergodic theory of the purely finitely additive part) is out of reach.
3. **Dynamics from observation:** “Can the Stone space give rise to dynamics without positing T ?” No: the algebra doesn’t constrain Ω , so it certainly doesn’t constrain structure on Ω .
4. **Paper III (noise thresholds):** Residual autocorrelation as embedding diagnostic = Billings–Voon [3] / Casdagli [8]. FNN noise sensitivity = Rhodes–Morari [33]. Elbow stopping = Lepski [27]. Rediscovery throughout.
5. **“No intermediate condition”:** The claim that no condition interpolates between first-order and σ -additivity is false. $\limsup \ell(A_n) < c$ trivially interpolates.
6. **Bridge theorem (fibre mixing):** The claimed equivalence $\delta(L) \rightarrow 0 \Leftrightarrow (\mu \otimes \mu)(R_L) \rightarrow 0$ under fibre mixing is false. The core identity has a disintegration error: the conditional variance integral is weighted by p_z , the collision integral by p_z^2 . The skew-product $X = A^{\mathbb{Z}} \times B^{\mathbb{Z}}$, $h(a, b) = a_0$ falsifies even the easy direction. The replacement observation (geometric $\neq$ algebraic reconstruction) is known to ergodic theorists. Both the positive bridge and the negative note fail the audit.

7. **Entropy characterization:** $\delta(L) \rightarrow 0 \Leftrightarrow H_2(\nu_L) \rightarrow \infty$ depended entirely on the bridge theorem. Dead.
8. **Nonlinear semigroups / Hille operator (Phase 1):** C_n -semigroups on Banach spaces, resolvent from finite difference spectra, max-entropy fitting. Abandoned approach — the semigroup formalism was too heavy for the applied question. The Rose kernel was the lighter successor.
9. **Rose–Takens measure conjugacy:** “Takens embedding + $D_{\text{KL}}(h_{\#}\mu\|\nu) = 0$ implies measure conjugacy.” Trivially true by definition: diffeomorphism + pushforward = target measure IS measure conjugacy. Not a theorem.
10. **CGF $\rightarrow$ Fisher $\rightarrow$ Landau chain:** Standard information geometry [2]. The CGF-Fisher connection is the definition of Fisher information.
11. **KL = least action (discrete Onsager–Machlup):** Known: [11], [28], large deviations literature.
12. **Fisher clock / entropy production:** Broken detailed balance $\rightarrow$ entropy production is [36]. The “Fisher clock” framing is poetic but the mathematics is standard nonequilibrium statistical mechanics.
13. **Neutral operator edge law:** The joint edge law $\Gamma = \nu \otimes K$ and its f -divergence structure were absorbed into Paper II (conditional regularity kernel κ_Q). The diagnostic exponents $\eta = 2s$, $\zeta = 2s/(2s + d)$ are the same Lepski/Stone rates already canned in Paper III.

7 What remains genuinely open

1. **OML extension problem:** Does every state on a general OML extend to a σ -additive measure on the McDonald–Bimbó dual $S_0(A)$? Gleason answers for $L(H)$; the general case is open. *Needs OML expertise to pursue.*
2. **Ergodic theory of purely finitely additive charges:** What are the extreme points of the purely finitely additive shift-invariant charges? Connected to Banach limits. *Out of reach currently.*

8 The understanding, condensed

One sentence per insight:

1. The passage from coherent charges to probability requires σ -additivity, which is not derivable from finitary conditions.
2. The Stone construction gives an unconditional measure on the dual space; descent to a realization requires a commitment the algebra cannot force.
3. Which ultrafilters are “realized” is additional structure in the Boolean case and constrained structure in the OML case.
4. Distributivity is sufficient for value-definiteness; non-distributivity blocks it for $L(H)$, $\dim \geq 3$, but not for all non-distributive OMLs (e.g. $L(\mathbb{C}^2)$, horizontal sums).
5. The realism/empiricism boundary is algebraic: van Fraassen’s empirical adequacy is the unconditional level; value-definite realism requires the algebra to admit dispersion-free states (guaranteed by distributivity, blocked by KS for $L(H)$, $\dim \geq 3$).

6. Delay reconstruction is algebraically underdetermined: embedding selection is necessarily empirical.
7. Geometric reconstruction ($\text{collision} \rightarrow 0$) and algebraic reconstruction ($\delta \rightarrow 0$) are genuinely inequivalent. The bridge theorem claiming their equivalence under fibre mixing was false (disintegration error in the core identity). The skew-product counterexample is definitive.
8. Every practical reconstruction diagnostic implicitly assumes a modelling commitment; noise violates deterministic commitments.
9. Audit before you draft.

9 References by subject

Measure extension and charges

- Halmos, *Measure Theory*, Van Nostrand, 1950. §13 (extension), §53–54 (Boolean algebras). [17]
- Billingsley, *Probability and Measure*, 3rd ed., Wiley, 1995. Ch. 36 (Kolmogorov extension). [4]
- Bogachev, *Measure Theory*, Vols. 1–2, Springer, 2007. Vol. 1 Ch. 1 (Carathéodory). [5]
- Rao & Rao, *Theory of Charges*, Academic Press, 1983. Thm 10.5.3 (Stone-space support characterization). [32]
- Yosida & Hewitt, “Finitely additive measures,” *Trans. AMS* 72 (1952), 46–66. [41]
- Choksi, “Inverse limits of measure spaces,” *Proc. London Math. Soc.* s3-8 (1958), 321–342. Thm 1. [9]
- Fremlin, *Measure Theory*, Vol. 3, Torres Fremlin, 2003. §326 (additive functionals on Boolean algebras). [13]

Stone duality

- Johnstone, *Stone Spaces*, Cambridge University Press, 1982. [22]

Quantum logic and contextuality

- Kalmbach, *Orthomodular Lattices*, Academic Press, 1983. Ch. 4 (states, valuations). [23]
- Pták & Pulmannová, *Orthomodular Structures as Quantum Logics*, Kluwer, 1991. [30]
- Kochen & Specker, “The problem of hidden variables in quantum mechanics,” *J. Math. Mech.* 17 (1967), 59–87. [25]
- Gleason, “Measures on the closed subspaces of a Hilbert space,” *J. Math. Mech.* 6 (1957), 885–893. [15]
- Bub & Clifton, “A uniqueness theorem for ‘no collapse’ interpretations,” *Stud. Hist. Phil. Mod. Phys.* 27 (1996), 181–219. [7]
- Halvorson & Clifton, “Maximal beable subalgebras,” *Int. J. Theor. Phys.* 38 (1999), 2441–2484. [18]

- Abramsky & Brandenburger, “The sheaf-theoretic structure of non-locality and contextuality,” *New J. Phys.* 13 (2011), 113036. [1]
- McDonald & Bimbó, “Topological duality for orthomodular lattices,” *Math. Logic Quart.* 69 (2023), 514–541. [29]
- Gunji, Haruna & Uragami, “Contextuality as a left adjoint,” arXiv:2603.22353 (2026). [16]

Topos approaches

- Isham & Butterfield, “Topos perspective on the Kochen–Specker theorem: I,” *Int. J. Theor. Phys.* 37 (1998), 2669–2733. [21]
- Heunen, Landsman & Spitters, “A topos for algebraic quantum theory,” *Comm. Math. Phys.* 291 (2009), 63–110. [19]

Delay embedding and reconstruction

- Takens, “Detecting strange attractors in turbulence,” *Lecture Notes in Math.* 898 (1981), 366–381. [39]
- Sauer, Yorke & Casdagli, “Embedology,” *J. Stat. Phys.* 65 (1991), 579–616. [35]
- Kennel, Brown & Abarbanel, “Determining embedding dimension for phase-space reconstruction,” *Phys. Rev. A* 45 (1992), 3403–3411. [24]
- Ragwitz & Kantz, “Markov models from data by simple nonlinear time series predictors,” *Phys. Rev. E* 65 (2002), 056201. [31]
- Botvinick-Greenhouse, Oprea & Maulik, “Measure-theoretic time-delay embedding,” *J. Stat. Phys.* 192 (2025), 171. [6]
- Rhodes & Morari, “False-nearest-neighbors algorithm and noise-corrupted time series,” *Phys. Rev. E* 55 (1997), 6162–6170. [33]

Ergodic theory

- Rokhlin, “Lectures on the entropy theory of measure-preserving transformations,” *Russian Math. Surveys* 22 (1967), 1–52. [34]
- Sinai, “On the concept of entropy of a dynamical system,” *Dokl. Akad. Nauk SSSR* 124 (1959), 768–771. [37]
- Krieger, “On entropy and generators of measure-preserving transformations,” *Trans. AMS* 149 (1970), 453–464. [26]
- Furstenberg, “Ergodic behavior of diagonal measures and a theorem of Szemerédi,” *J. Analyse Math.* 31 (1977), 204–256. [14]

Philosophy of science

- van Fraassen, *The Scientific Image*, Oxford University Press, 1980. [40]
- de Finetti, *Theory of Probability*, Vol. 1, Wiley, 1974. [10]

Nonparametric statistics

- Stone, “Optimal global rates of convergence for nonparametric regression,” *Ann. Statist.* 10 (1982), 1040–1053. [38]
- Lepski, “On a problem of adaptive estimation in Gaussian white noise,” *Theory Probab. Appl.* 35 (1991), 454–466. [27]
- Billings & Voon, “Correlation based model validity tests for non-linear models,” *Int. J. Control* 44 (1986), 235–244. [3]
- Casdagli, “Chaos and deterministic versus stochastic non-linear modelling,” *J. Roy. Statist. Soc. B* 54 (1992), 303–328. [8]

Model theory and probability logic

- Hoover, “Probability logic,” *Ann. Math. Logic* 14 (1978), 287–313. [20]
- Fagin, Halpern & Megiddo, “A logic for reasoning about probabilities,” *Inform. Comput.* 87 (1990), 78–128. [12]

References

- [1] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New J. Phys.*, 13:113036, 2011.
- [2] Shun-ichi Amari. *Differential-Geometrical Methods in Statistics*. Springer, 1985.
- [3] Stephen A. Billings and W. S. F. Voon. Correlation based model validity tests for non-linear models. *Int. J. Control*, 44:235–244, 1986.
- [4] Patrick Billingsley. *Probability and Measure*. Wiley, 3rd edition, 1995.
- [5] Vladimir I. Bogachev. *Measure Theory*, volume 1–2. Springer, 2007.
- [6] J. Botvinick-Greenhouse, M. Oprea, and R. Maulik. Measure-theoretic time-delay embedding. *J. Stat. Phys.*, 192:171, 2025.
- [7] Jeffrey Bub and Rob Clifton. A uniqueness theorem for ‘no collapse’ interpretations of quantum mechanics. *Stud. Hist. Phil. Mod. Phys.*, 27:181–219, 1996.
- [8] Martin Casdagli. Chaos and deterministic versus stochastic non-linear modelling. *J. Roy. Statist. Soc. B*, 54:303–328, 1992.
- [9] J. R. Choksi. Inverse limits of measure spaces. *Proc. London Math. Soc.*, s3-8:321–342, 1958.
- [10] Bruno de Finetti. *Theory of Probability*, volume 1. Wiley, 1974.
- [11] Gregory L. Eyink. Action principle in nonequilibrium statistical dynamics. *Phys. Rev. E*, 54:3419–3435, 1996.
- [12] Ronald Fagin, Joseph Y. Halpern, and Nimrod Megiddo. A logic for reasoning about probabilities. *Inform. Comput.*, 87:78–128, 1990.
- [13] David H. Fremlin. *Measure Theory*, volume 3. Torres Fremlin, 2003.

- [14] Harry Furstenberg. Ergodic behavior of diagonal measures and a theorem of Szemerédi. *J. Analyse Math.*, 31:204–256, 1977.
- [15] Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *J. Math. Mech.*, 6:885–893, 1957.
- [16] Yukio-Pegio Gunji, Taichi Haruna, and Daisuke Urugami. Contextuality as a left adjoint. *arXiv:2603.22353*, 2026.
- [17] Paul R. Halmos. *Measure Theory*. Van Nostrand, 1950.
- [18] Hans Halvorson and Rob Clifton. Maximal beable subalgebras of quantum-mechanical observables. *Int. J. Theor. Phys.*, 38:2441–2484, 1999.
- [19] Chris Heunen, Nicolaas P. Landsman, and Bas Spitters. A topos for algebraic quantum theory. *Comm. Math. Phys.*, 291:63–110, 2009.
- [20] Douglas N. Hoover. Probability logic. *Ann. Math. Logic*, 14:287–313, 1978.
- [21] Christopher J. Isham and Jeremy Butterfield. Topos perspective on the Kochen–Specker theorem: I. *Int. J. Theor. Phys.*, 37:2669–2733, 1998.
- [22] Peter T. Johnstone. *Stone Spaces*. Cambridge University Press, 1982.
- [23] Gudrun Kalmbach. *Orthomodular Lattices*. Academic Press, 1983.
- [24] Matthew B. Kennel, Reggie Brown, and Henry D. I. Abarbanel. Determining embedding dimension for phase-space reconstruction. *Phys. Rev. A*, 45:3403–3411, 1992.
- [25] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *J. Math. Mech.*, 17:59–87, 1967.
- [26] Wolfgang Krieger. On entropy and generators of measure-preserving transformations. *Trans. Amer. Math. Soc.*, 149:453–464, 1970.
- [27] Oleg V. Lepski. On a problem of adaptive estimation in Gaussian white noise. *Theory Probab. Appl.*, 35:454–466, 1991.
- [28] Christian Maes and Karel Netočný. Time-reversal and entropy. *J. Stat. Phys.*, 110:269–310, 2003.
- [29] Joseph McDonald and Katalin Bimbó. Topological duality for orthomodular lattices. *Math. Logic Quart.*, 69:514–541, 2023.
- [30] Pavel Pták and Sylvia Pulmannová. *Orthomodular Structures as Quantum Logics*. Kluwer, 1991.
- [31] Mario Ragwitz and Holger Kantz. Markov models from data by simple nonlinear time series predictors. *Phys. Rev. E*, 65:056201, 2002.
- [32] K. P. S. Bhaskara Rao and M. Bhaskara Rao. *Theory of Charges*. Academic Press, 1983.
- [33] Carl Rhodes and Manfred Morari. False-nearest-neighbors algorithm and noise-corrupted time series. *Phys. Rev. E*, 55:6162–6170, 1997.
- [34] V. A. Rokhlin. Lectures on the entropy theory of measure-preserving transformations. *Russian Math. Surveys*, 22:1–52, 1967.

- [35] Tim Sauer, James A. Yorke, and Martin Casdagli. Embedology. *J. Stat. Phys.*, 65:579–616, 1991.
- [36] Jürgen Schnakenberg. Network theory of microscopic and macroscopic behavior of master equation systems. *Rev. Mod. Phys.*, 48:571–585, 1976.
- [37] Ya. G. Sinai. On the concept of entropy of a dynamical system. *Dokl. Akad. Nauk SSSR*, 124:768–771, 1959.
- [38] Charles J. Stone. Optimal global rates of convergence for nonparametric regression. *Ann. Statist.*, 10:1040–1053, 1982.
- [39] Floris Takens. Detecting strange attractors in turbulence. *Lecture Notes in Math.*, 898:366–381, 1981.
- [40] Bas C. van Fraassen. *The Scientific Image*. Oxford University Press, 1980.
- [41] Kōsaku Yosida and Edwin Hewitt. Finitely additive measures. *Trans. Amer. Math. Soc.*, 72:46–66, 1952.